

CV Date

03/03/2026

Part A. PERSONAL INFORMATION

First Name *	Umesh		
Family Name *	Kumar		
Sex *	Male	Date of Birth *	01/02/1991
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Researcher's identification number	Open Researcher and Contributor ID (ORCID) *	0000-0002-0961-8048	
	Researcher ID	ABD-9355-2020	
	Scopus Author ID	57208040930	

* Mandatory

A.1. Current position

Job Title	Guest Researcher / Lecturer		
Starting date	2024		
Institution	Chhatrapati Shahu Ji Maharaj University		
Department / Centre	Department of Nephrology / Sanjay Gandhi Postgraduate Institute of Medical Sciences		
Country	India	Phone Number	
Keywords			

A.2. Previous positions

Period	Job Title / Name of Employer / Country
2023 - 2024	Postdoctoral Researcher / Leopold-Franzens Universität Innsbruck / Austria
2021 - 2023	Senior Research Associate / National Bureau of Animal Genetic Resources / India
2018 - 2021	Predocctoral Researcher (Senior Research Fellow - SRF) / Centre of Biomedical Research (CBMR) / India
2015 - 2018	Predocctoral Researcher (Junior Research Fellow - JRF) / Centre of Biomedical Research (CBMR) / India
2014 - 2015	Project Assistant / Bareilly College / India
2012 - 2013	Business Development Executive / GenxBio Health Sciences pv. Ltd. / India
2011 - 2011	Summer Trainee / Indian Institute of Sciences / India

A.3. Education

Degree/Master/PhD	University / Country	Year
PhD in Clinical Metabolomics	Babasaheb Bhimrao Ambedkar University / India	2022
Master of Science	Babasaheb Bhimrao Ambedkar University / India	2012
Bachelor of Science in Chemistry	Mahatama Jyotiba Phule Rohilkhand University / India	2010

Part B. CV SUMMARY

Academic and Professional Background: I hold an MSc in Biological Sciences and have qualified four competitive national-level examinations in India: GATE (2014), ICMR–Junior

Research Fellowship (2014), CSIR–NET Lectureship (2015), and the UPHESC Assistant Professorship Examination (2022), securing top national ranks. I initiated my research career as a Junior Research Fellow (2015–2018) at Centre of Biomedical Research (India), supported by an ICMR research grant. I subsequently completed my PhD in Clinical Metabolomics (2018–2022) at Babasaheb Bhimrao Ambedkar University (India), funded by the ICMR–Senior Research Fellowship scheme. Following my doctoral training, I joined the National Bureau of Animal Genetic Resources (India), in 2022 as a Research Associate-III on a DST-India funded national project, where I assumed senior scientific and mentoring responsibilities. In May 2023, I moved to Europe and worked as a Postdoctoral Researcher at the University of Innsbruck, Austria (May 2023–June 2024), within Bruker- and VASCa-funded research programs, gaining independent project leadership experience in a highly interdisciplinary and international environment.

Scientific Activity, Research Profile, and Leadership: My research activity focuses on metabolomics-driven biomarker discovery using NMR, LC-MS/MS, and GC-MS platforms, with applications in precision medicine, early disease diagnosis and prognosis, biomedicine, nutrition, gut microbiota, and vascular and neonatal health. I have extensive experience in SOP development, large-scale clinical and population-based sample processing, multivariate statistical analysis, and translational interpretation of metabolic signatures. During my postdoctoral tenure at the University of Innsbruck, both of my supervisors retired, after which I independently led and completed my assigned research project, resulting in a high-quality manuscript. In parallel, I established and sustained international collaborations with the European Translational Oncology Prevention and Screening Institute (EUTOPS, Austria) within the TIROLGESUND ATLAS project and with the Medical University of Innsbruck under the NEOVASCa project, investigating antibiotic-associated metabolic alterations in neonates. Earlier in my career, I coordinated collaborative research projects with the Departments of Gastroenterology and Immunology at SGPGIMS Hospital (India) and with the Pharmaceutical Sciences and Zoology Departments of BBAU (Central University, India), contributing to clinically relevant and interdisciplinary translational research.

Supervision, Mentoring, and Academic Responsibilities: I have made sustained contributions to research supervision and capacity building. During my doctoral training, I mentored 2 PhD scholars and 5 MSc students in metabolomics-based dissertation research. As a Research Associate at NBAGR, I held a senior mentoring role, supervising 1 Senior Research Fellow, 3 PhD scholars, and 1 Master's student within my principal investigator's group. In addition to research leadership, I have held significant academic administrative responsibilities, serving three times as Examination Centre In-Charge Officer for undergraduate and postgraduate examinations conducted by CSJM University, with responsibility for institutional-level examination coordination and compliance.

Teaching, Funding, and Scientific Engagement: From September 2024 to December 2025, I have been serving as Guest Faculty at a CSJM-University affiliated college in India, teaching Human Physiology, Developmental Biology, and Biochemistry to BSc and MSc students, and mentoring students in laboratory practicals and dissertation projects. I have received ICMR travel and workshop grants to attend national and international conferences, seminars, and workshops. I am a Founder Member of the Translational Biomedical Research Society, India, and a life member of NMRS, TBRIS, ISETF, and APNMR. My scientific engagement includes 2 invited lectures, 7 oral and poster presentations, and participation in multiple national and international scientific events, where I have also received best oral and poster presentation awards.

Scientific Output, Editorial Activity, and Impact: I have published >50 peer-reviewed international scientific articles (1120 citations, h-index 20). My publication portfolio includes 21 Q1 and 20 Q2 papers, with 12 publications in top 10 percentile, demonstrating sustained productivity, leadership in collaborative research, and strong international visibility. My authorship includes 9 publications as 1st author, 14 as 2nd author, 7 as 3rd author, and the remaining as contributing co-author. I have actively contributed to scientific quality assurance and peer review, and currently serve as a Reviewing Editor for Molecular Diagnostics & Therapeutics, thus reflecting sustained engagement in editorial and peer-review activities across biomedical and translational research domains.

Part C. RELEVANT ACCOMPLISHMENTS

C.1. Publications

AC: corresponding author. (n° x / n° y): position / total authors. If applicable, indicate the number of citations

- 1 **Scientific paper.** Vidhi Garg; Manishi Mukesh; (3/9) UMESH KUMAR; et al; Monika Sodhi. 2025. Characterization of metabolite profiles in milk derived exosomes from indicus, crossbred and taurine cows by proton nuclear magnetic resonance analysis. Food Chemistry. ELSEVIER SCI LTD. 473, pp.143015. ISSN 0308-8146. <https://doi.org/10.1016/j.foodchem.2025.143015>
- 2 **Scientific paper.** Amarjeet; (2/17) UMESH KUMAR; Monika Sodhi; et al; Manishi Mukesh. 2025. Characterizing metabolome signature of colostrum, transition and mature milk of indigenous cows (Bos indicus) adapted to high altitude environment of Leh-Ladakh. Food Chemistry. ELSEVIER SCI LTD. 464, pp.141767. ISSN 0308-8146. <https://doi.org/10.1016/j.foodchem.2024.141767>
- 3 **Scientific paper.** Chiara Maria Stella Herzog; Charlotte D. Vavourakis; Bente Theeuwes; et al; Martin Widschwendter; (10/25) UMESH KUMAR. 2025. Longitudinal multi-omic evaluation of biomarkers of health and ageing over smoking cessation intervention. bioRxiv. ISSN 2692-8205. <https://doi.org/10.1101/2025.08.29.673135>
- 4 **Scientific paper.** Chiara Maria Stella Herzog; Charlotte D. Vavourakis; Elisa Redl; et al; Martin Widschwendter; (7/28) UMESH KUMAR. 2025. Multi-modal atlas of lifestyle interventions reveals malleability of ageing-linked molecular features. bioRxiv. ISSN 2692-8205. <https://doi.org/10.1101/2025.08.30.673115>
- 5 **Scientific paper.** Chiara Maria Stella Herzog; Charlotte D. Vavourakis; Bente Theeuwes; et al; Martin Widschwendter; (10/26) UMESH KUMAR. 2025. Systemic multi-omic remodelling underlies health benefits of intermittent fasting. bioRxiv. ISSN 2692-8205. <https://doi.org/10.1101/2025.08.30.673138>
- 6 **Scientific paper.** Samagra Agarwal; Pooja Goswami; Shekhar Poudel; et al; Anoop Saraya; (7/9) UMESH KUMAR. 2023. Acute pancreatitis is characterized by generalized intestinal barrier dysfunction in early stage. Pancreatology. Elsevier. 23-1, pp.9-17. ISSN 1424-3903. <https://doi.org/10.1016/j.pan.2022.11.011>
- 7 **Scientific paper.** Anusmita Shekher; Puneet; Nikee Awasthee; (4/7) UMESH KUMAR; Ritu Raj; Dinesh Kumar; Subash Chandra Gupta. 2023. Association of altered metabolic profiles and long non-coding RNAs expression with disease severity in breast cancer patients: analysis by ¹H NMR spectroscopy and RT-q-PCR. Metabolomics. Springer Nature. 19, pp.8. ISSN 1573-3882. <https://doi.org/10.1007/s11306-023-01972-5>
- 8 **Scientific paper.** Venkatesh Kumar Ramappa; Vandana Singh; Devika Srivastava; et al; Dinesh Kumar; (9/10) UMESH KUMAR. 2023. Fabrication of mulberry leaf extract (MLE)- and tasar pupal oil (TPO)-loaded silk fibroin (SF) hydrogels and their antimicrobial properties. 3 Biotech. Springer. 13, pp.37. ISSN 2190-572X. <https://doi.org/10.1007/s13205-022-03443-5>
- 9 **Scientific paper.** Vandana Singh; Deepak Kumar Tripathi; Vivek Kumar Sharma; et al; Venkatesh Kumar; (5/9) UMESH KUMAR. 2023. Silk fibroin hydrogel: A novel biopolymer for sustained release of vancomycin drug for diabetic wound healing. Journal of Molecular Structure. Elsevier. 1286, pp.135548. ISSN 0022-2860. <https://doi.org/10.1016/j.molstruc.2023.135548>
- 10 **Scientific paper.** Archana Bharti Sonkar; Pranesh Kumar; Anand Kumar; et al; Sudipta Saha; (7/13) UMESH KUMAR. 2023. Vinpocetine mitigates DMH-induce pre-neoplastic colon damage in rats through inhibition of pro-inflammatory cytokines. International Immunopharmacology. Elsevier. 119, pp.110236. ISSN 1567-5769. <https://doi.org/10.1016/j.intimp.2023.110236>

- 11 Scientific paper.** Pranesh Kumar; Mohit Kumar; Anurag Kumar Gautam; et al; Sudipta Saha; (8/15) UMESH KUMAR. 2022. Ameliorative effect of fluvoxamine against colon carcinogenesis via COX-2 blockade with oxidative and metabolic stress reduction at the cellular, molecular and metabolic levels. BBA Advances. Elsevier. 2, pp.100046. ISSN 2667-1603. <https://doi.org/10.1016/j.bbadv.2022.100046>
- 12 Scientific paper.** Raquibun Nisha; Pranesh Kumar; (3/11) UMESH KUMAR; et al; Shubhini A. Saraf. 2022. Assessment of hyaluronic acid-modified imatinib mesylate cubosomes through CD44 targeted drug delivery in NDEA-induced hepatic carcinoma. International Journal of Pharmaceutics. Elsevier. 622, pp.121848. ISSN 0378-5173. <https://doi.org/10.1016/j.ijpharm.2022.121848>
- 13 Scientific paper.** Pranesh Kumar; Anurag Kumar Gautam; (3/10) UMESH KUMAR; et al; Sudipta Saha. 2022. Mechanistic exploration of the activities of poly(lactic-co-glycolic acid)-loaded nanoparticles of betulinic acid against hepatocellular carcinoma at cellular and molecular levels. Archives of Physiology and Biochemistry. TAYLOR & FRANCIS LTD. 128-3, pp.836-848. ISSN 1381-3455. <https://doi.org/10.1080/13813455.2020.1733024>
- 14 Scientific paper.** Gaurav Pande; Manjunath Hatti; Mohit Kumar Rai; et al; Vikas Agarwal; (12/19) UMESH KUMAR. 2022. Response Guided Slow Infusion of Albumin, Vasoconstrictors and Furosemide Improves Ascites Mobilization and Survival in Acute on Chronic Liver Failure: A Proof-of-Concept Study. Journal of Inflammation Research. DOVE MEDICAL PRESS LTD. 2022-15, pp.5027-5039. ISSN 1178-7031. <https://doi.org/10.2147/JIR.S377494>
- 15 Scientific paper.** (1/8) UMESH KUMAR; Abhai Kumar; Smita Singh; Payal Arya; Sandeep Kumar Singh; Rameshwar Nath Chaurasia; Anup Singh; Dinesh Kumar. 2021. An elaborative NMR based plasma metabolomics study revealed metabolic derangements in patients with mild cognitive impairment: a study on north Indian population. Metabolic Brain Disease. Springer. 36, pp.957-968. ISSN 0885-7490. <https://doi.org/10.1007/s11011-021-00700-z>
- 16 Scientific paper.** (1/8) UMESH KUMAR; Pankti Mehta; Sandeep Kumar; Avinash Jain; Anupam Guleria; Venkatesh Kumar R; Ramnath Misra; Dinesh Kumar. 2021. Circulatory histidine levels as predictive indicators of disease activity in takayasu arteritis. Analytical Science Advances. Wiley. 2-11-12, pp.527-535. ISSN 2628-5452. <https://doi.org/10.1002/ansa.202000181>
- 17 Scientific paper.** Raquibun Nisha; Pranesh Kumar; (3/10) UMESH KUMAR; et al; Shubhini A Saraf. 2021. Fabrication of Imatinib Mesylate-Loaded Lactoferrin-Modified PEGylated Liquid Crystalline Nanoparticles for Mitochondrial-Dependent Apoptosis in Hepatocellular Carcinoma. Molecular Pharmaceutics. ACS. 18-3, pp.1102-1120. ISSN 1543-8384. <https://doi.org/10.1021/acs.molpharmaceut.0c01024>
- 18 Scientific paper.** Anupam Guleria; (2/10) UMESH KUMAR; Dinesh Kumar; et al; Latika Gupta. 2021. NMR-based serum and muscle metabolomics for diagnosis and activity assessment in idiopathic inflammatory myopathies. Analytical Science Advances. Wiley. 2-11-12, pp.515-526. ISSN 2628-5452. <https://doi.org/10.1002/ansa.202000171>
- 19 Scientific paper.** (1/13) UMESH KUMAR; Supriya Sharma; Manjunath Durgappa; et al; Dinesh Kumar. 2021. Serum Metabolic Disturbances Associated with Acute-on-chronic Liver Failure in Patients with Underlying Alcoholic Liver Diseases: An Elaborative NMR-based Metabolomics Study. Journal of Pharmacy And Bioallied Sciences. WOLTERS KLUWER MEDKNOW PUBLICATIONS. 13-2, pp.276-282. ISSN 0976-4879. https://doi.org/10.4103/JPBS.JPBS_333_20
- 20 Scientific paper.** Payal Arya; (2/8) UMESH KUMAR; Supriya Sharma; Manjunath Durgappa; Anupam Guleria; Ritu Raj; Gaurav Pande; Dinesh Kumar. 2021. Targeted NMR-based serum metabolic profiling of serine, glycine and methionine in acute-on-chronic liver failure patients: Possible insights into mitochondrial dysfunction. Analytical Science Advances. Wiley. 2-11-12, pp.536-545. ISSN 2628-5452. <https://doi.org/10.1002/ansa.202000167>

- 21 Scientific paper.** R. Venkatesh Kumar; Devika Srivastava; Vandana Singh; (4/8) UMESH KUMAR; Vijay Kumar Vishvakarma; Prashant Singh; Dinesh Kumar; Rajesh Kumar. 2020. Characterization, biological evaluation and molecular docking of mulberry fruit pectin. *Scientific Reports. Nature.* 10, pp.21789. ISSN 2045-2322. <https://doi.org/10.1038/s41598-020-78086-8>
- 22 Scientific paper.** (1/9) UMESH KUMAR; Avinash Jain; Anupam Guleria; et al; Dinesh Kumar. 2020. Circulatory Glutamine/Glucose ratio for evaluating disease activity in Takayasu arteritis: A NMR based serum metabolomics study. *Journal of Pharmaceutical and Biomedical Analysis. Elsevier.* 180, pp.113080. ISSN 0731-7085. <https://doi.org/10.1016/j.jpba.2019.113080>
- 23 Scientific paper.** Sujata Ganguly; (2/11) UMESH KUMAR; Nikhil Gupta; et al; Ramnath Misra. 2020. Nuclear magnetic resonance-based targeted profiling of urinary acetate and citrate following cyclophosphamide therapy in patients with lupus nephritis. *Lupus. Sage Publications LTD.* 29-7, pp.782-786. ISSN 0961-2033. <https://doi.org/10.1177/0961203320918011>
- 24 Scientific paper.** Pranesh Kumar; Aakriti Agarwal; Ashok K. Singh; et al; Sudipta Saha; (6/13) UMESH KUMAR. 2019. Antineoplastic properties of zafirlukast against hepatocellular carcinoma via activation of mitochondrial mediated apoptosis. *Regulatory Toxicology and Pharmacology. Elsevier.* 109, pp.104489. ISSN 0273-2300. <https://doi.org/10.1016/j.yrtph.2019.104489>
- 25 Scientific paper.** Vinit Raj; Archana S Bhadauria; Ashok K Singh; et al; Sudipta Saha; (4/14) UMESH KUMAR. 2019. Novel 1,3,4-thiadiazoles inhibit colorectal cancer via blockade of IL-6/COX-2 mediated JAK2/STAT3 signals as evidenced through data-based mathematical modeling. *Cytokine. Elsevier.* 118, pp.144-159. ISSN 1043-4666. <https://doi.org/10.1016/j.cyto.2018.03.026>
- 26 Scientific paper.** Vimal Maurya; Pranesh Kumar; Sreemoyee Chakraborti; et al; Sudipta Saha; (6/14) UMESH KUMAR. 2019. Zolmitriptan attenuates hepatocellular carcinoma via activation of caspase mediated apoptosis. *Chemico-Biological Interactions. Elsevier.* 308, pp.120-129. ISSN 0009-2797. <https://doi.org/10.1016/j.cbi.2019.05.033>
- 27 Scientific paper.** Priyanka Mishra; Vinit Raj; Archana S Bhadauria; et al; Sudipta Saha; (10/16) UMESH KUMAR. 2018. 6,7-dimethoxy-1,2,3,4-tetrahydro-isoquinoline-3-carboxylic acid attenuates colon carcinogenesis via blockade of IL-6 mediated signals. *Biomedicine & Pharmacotherapy. Elsevier.* 100, pp.282-295. ISSN 0753-3322. <https://doi.org/10.1016/j.biopha.2018.02.009>
- 28 Scientific paper.** Vinit Raj; Amit Rai; Ashok K Singh; et al; Sudipta Saha; (7/9) UMESH KUMAR. 2018. Discovery of Novel 2-Amino-5-(Substituted)-1,3,4-Thiadiazole Derivatives: New Utilities for Colon Cancer Treatment. *Anti-Cancer Agents in Medicinal Chemistry. Bentham Science.* 18-5, pp.719-738. ISSN 1871-5206. <https://doi.org/10.2174/1871520617666170419122916>
- 29 Scientific paper.** A Rawat; S Chaturvedi; A K Singh; et al; S Saha; (10/12) U KUMAR. 2018. Metabolomics approach discriminates toxicity index of pyrazinamide and its metabolic products, pyrazinoic acid and 5-hydroxy pyrazinoic acid. *Human & Experimental Toxicology. Sage Publications LTD.* 37-4, pp.373-389. ISSN 0960-3271. <https://doi.org/10.1177/0960327117705426>
- 30 Scientific paper.** Anupam Guleria; Sanat Phatak; Durgesh Dubey; et al; Ramnath Misra; (7/11) UMESH KUMAR. 2018. NMR-Based Serum Metabolomics Reveals Reprogramming of Lipid Dysregulation Following Cyclophosphamide-Based Induction Therapy in Lupus Nephritis. *Journal of Proteome Research. ACS.* 17-7, pp.2440-2448. ISSN 1535-3893. <https://doi.org/10.1021/acs.jproteome.8b00192>
- 31 Scientific paper.** Avinash Jain; Dinesh Kumar; Anupam Guleria; et al; Ramnath Misra; (8/12) UMESH KUMAR. 2018. NMR-Based Serum Metabolomics of Patients with Takayasu Arteritis: Relationship with Disease Activity. *Journal of Proteome Research. ACS.* 17-9, pp.3317-3324. ISSN 1535-3893. <https://doi.org/10.1021/acs.jproteome.8b00456>

- 32 **Scientific paper.** Amit Rai; (2/15) UMESH KUMAR; Vinit Raj; et al; Sudipta Saha. 2018. Novel 1,4-benzothiazines obliterate COX-2 mediated JAK-2/STAT-3 signals with potential regulation of oxidative and metabolic stress during colorectal cancer. *Pharmacological Research*. Elsevier. 132, pp.188-203. ISSN 1043-6618. <https://doi.org/10.1016/j.phrs.2017.12.010>
- 33 **Scientific paper.** Ashok K. Singh; Archana S. Bhadauria; (3/12) UMESH KUMAR; et al; Sudipta Saha. 2018. Novel Indole-fused benzo-oxazepines (IFBOs) inhibit invasion of hepatocellular carcinoma by targeting IL-6 mediated JAK2/STAT3 oncogenic signals. *Scientific Reports*. Nature. 8, pp.5932. ISSN 2045-2322. <https://doi.org/10.1038/s41598-018-24288-0>
- 34 **Scientific paper.** Ashok K Singh; Archana Singh Bhadauria; (3/11) UMESH KUMAR; et al; Sudipta Saha. 2018. Novel fused oxazepino-indoles (FOIs) attenuate liver carcinogenesis via IL-6/JAK2/STAT3 signaling blockade as evidenced through data-based mathematical modeling. *Life Sciences*. Elsevier. 201, pp.161-172. ISSN 0024-3205. <https://doi.org/10.1016/j.lfs.2018.02.029>
- 35 **Scientific paper.** Amit K Keshari; Ashok K Singh; (3/11) UMESH KUMAR; et al; Sudipta Saha. 2017. 5H-benzo[h]thiazolo[2,3-b]quinazolines ameliorate NDEA-induced hepatocellular carcinogenesis in rats through IL-6 downregulation along with oxidative and metabolic stress reduction. *Drug Design, Development and Therapy*. DOVE Medical Press. 11, pp.2981-2995. ISSN 1177-8881. <https://doi.org/10.2147/DDDT.S143075>
- 36 **Scientific paper.** Pranesh Kumar; Ashok K Singh; Vinit Raj; et al; Sudipta Saha; (7/11) UMESH KUMAR. 2017. 6,7-dimethoxy-1,2,3,4-tetrahydro-isoquinoline-3-carboxylic acid attenuates hepatocellular carcinoma in rats with NMR-based metabolic perturbations. *Future Science OA*. Taylor & Francis. 3-3, pp.FSO202. ISSN 2056-5623. <https://doi.org/10.4155/fsoa-2017-0008>
- 37 **Scientific paper.** Anil K Sahdev; Vinit Raj; Ashok K Singh; et al; Sudipta Saha; (8/13) UMESH KUMAR. 2017. Ameliorative effects of pyrazinoic acid against oxidative and metabolic stress manifested in rats with dimethylhydrazine induced colonic carcinoma. *Cancer Biology & Therapy*. Taylor & Francis. 18-5, pp.304-313. ISSN 1538-4047. <https://doi.org/10.1080/15384047.2017.1310341>
- 38 **Scientific paper.** Amit Rai; Vinit Raj; Ashok K. Singh; Amit K. Keshari; (5/7) UMESH KUMAR; Dinesh Kumar; Sudipta Saha. 2017. Design and synthesis of 1,4-benzothiazine derivatives with promising effects against colorectal cancer cells. *Cogent Chemistry*. Taylor & Francis. 3-1, pp.1303909. ISSN 2331-2009. <https://doi.org/10.1080/23312009.2017.1303909>
- 39 **Scientific paper.** Dinesh Kumar; Gaurav Pandey; Deepak Bansal; Atul Rawat; (5/8) UMESH KUMAR; Durgesh Dubey; Anupam Guleria; Vivek Anand Saraswat. 2017. NMR-based urinary profiling of lactulose/mannitol ratio used to assess the altered intestinal permeability in acute on chronic liver failure (ACLF) patients. *Magnetic Resonance in Chemistry*. Wiley. 55-4, pp.289-296. ISSN 0749-1581. <https://doi.org/10.1002/mrc.4525>
- 40 **Scientific paper.** Amit K Keshari; Ashok K Singh; Vinit Raj; et al; Sudipta Saha; (7/10) UMESH KUMAR. 2017. p-TSA-promoted syntheses of 5H-benzo[h]thiazolo[2,3-b]quinazoline and indeno[1,2-d]thiazolo[3,2-a]pyrimidine analogs: molecular modeling and in vitro antitumor activity against hepatocellular carcinoma. *Drug Design, Development and Therapy*. DOVE Medical Press. 11, pp.1623-1642. ISSN 1177-8881. <https://doi.org/10.2147/DDDT.S136692>
- 41 **Scientific paper.** Atul Rawat; Durgesh Dubey; Anupam Guleria; et al; Dinesh Kumar; (4/9) UMESH KUMAR. 2016. 1H NMR-based serum metabolomics reveals erythromycin-induced liver toxicity in albino Wistar rats. *Journal of Pharmacy and Bioallied Sciences*. Wolters Kluwer. 8-4, pp.327-334. ISSN 0976-4879. <https://doi.org/10.4103/0975-7406.199339>
- 42 **Scientific paper.** Anupam Guleria; Avadhesh Pratap; Durgesh Dubey; et al; Dinesh Kumar; (9/13) UMESH KUMAR. 2016. NMR based serum metabolomics reveals a distinctive signature in patients with Lupus Nephritis. *Scientific Reports*. Nature. 6, pp.35309. ISSN 2045-2322. <https://doi.org/10.1038/srep35309>

- 43 Review.** Venkatesh Kumar; Devika Srivastava; Abhishek Verma; Vandana Singh; (5/5) UMESH KUMAR. 2026. Kuwanons from Mulberry (*Morus* spp.): Exploring their structural activity relationship, biosynthesis, and biological properties. *Fitoterapia*. Elsevier. 190, pp.107138. ISSN 0367-326X. <https://doi.org/10.1016/j.fitote.2026.107138>
- 44 Review.** Venkatesh Kumar; Wouatsa Nangue Arlette Vyry; Seema Chauhan; Vandana Singh; Devika Srivastava; (6/8) UMESH KUMAR; Ritu Raj; Abhishek Verma. 2024. Mulberry: From nutraceuticals to bioactive phytochemicals. *Food and Humanity*. Elsevier. 2, pp.100272. ISSN 2949-8244. <https://doi.org/10.1016/j.foohum.2024.100272>
- 45 Review.** Sandeep Kumar Singh; (2/4) UMESH KUMAR; Anupam Guleria; Dinesh Kumar. 2021. A brief overview about the use of different bioactive liposome-based drug delivery systems in Peritoneal Dialysis and some other diseases. *Nano Express*. IOP Publishing Ltd. 2-2, pp.022006. ISSN 2632-959X. <https://doi.org/10.1088/2632-959X/abfdd1>
- 46 Review.** Devika Srivastava; Vandana Singh; (3/4) UMESH KUMAR; Venkatesh Kumar R. 2021. Alpha-Linolenic acid: A Pharmacologically Active Ingredient from Nature. *Indian Journal of Nutrition and Dietetics*. Informatics Publishing Limited. 58-4, pp.534-553. ISSN 0022-3174. <https://doi.org/10.21048/IJND.2021.58.4.28086>
- 47 Review.** R. Venkatesh Kumar; Devika Srivastava; (3/5) UMESH KUMAR; Mukesh Kumar; Prashant Singh. 2020. Bioprospecting of Omega 3 Fatty Acid from Silkworm Pupal Oil: From Molecular Mechanism to Biological Activities. *Journal of Biologically Active Products from Nature*. Taylor & Francis. 10-6, pp.495-506. ISSN 2231-1866. <https://doi.org/10.1080/22311866.2020.1862704>
- 48 Review.** Venkatesh Kumar Ramappa; Devika Srivastava; Prashant Singh; et al; Ritu Raj; (4/9) UMESH KUMAR. 2020. Mulberries: A Promising Fruit for Phytochemicals, Nutraceuticals, and Biological Activities. *International Journal of Fruit Science*. Taylor & Francis. 20-3, pp.1254-1279. ISSN 1553-8362. <https://doi.org/10.1080/15538362.2020.1784075>
- 49 Review.** Venkatesh Kumar Ramappa; Devika Srivastava; Prashant Singh; (4/5) UMESH KUMAR; Vandana Singh. 2020. Mulberry 1-deoxynojirimycin (DNJ): an exemplary compound for therapeutics. *Journal of Horticultural Science and Biotechnology*. Taylor & Francis. 95-6, pp.679-686. ISSN 1462-0316. <https://doi.org/10.1080/14620316.2020.1760738>
- 50 Review.** Anupam Guleria; Amit Kumar; (3/5) UMESH KUMAR; Ritu Raj; Dinesh Kumar. 2018. NMR Based Metabolomics: An Exquisite and Facile Method for Evaluating Therapeutic Efficacy and Screening Drug Toxicity. *Current Topics in Medicinal Chemistry*. Bentham Science. 18-20, pp.1827-1849. ISSN 1568-0266. <https://doi.org/10.2174/1568026619666181120141603>

C.3. Research projects and contracts

- 1 Project.** SEED/ASACODER-076/2018, Characterizing Milk Colostrum of Ladakhi Cows and Yak for identification of Biomolecules with Therapeutic Potential. Department of Science and Technology. Manishi Mukesh. (National Bureau of Animal Genetic Resources). 28/12/2020-27/12/2023. 230.725 €. Team member. Sample Collection, Sample preparation, Coordination with collaborative institute, supervised junior researchers, Data collection, preparation of daily strategies for project work, Manuscript preparat...
- 2 Project.** 3/1/3/JRF-2014/HRD, Identification of distinctive metabolic signature of autoimmune inflamma-tory diseases using NMR based metabolomics approach. Indian Council of Medical Research. Umesh Kumar. (Centre of Biomedical Research). 31/08/2018-30/07/2020. 9.700 €. Principal investigator. Conceptualization, study design, sample collection, sample preparation, Data recording and analysis, Manuscript preparation, project report

- 3 Project.** 3/1/3/JRF-2014/HRD, Metabolic disturbances associated with autoimmune diseases: Insights into disease mechanism and disease finger printing using NMR based metabolomics. Indian Council of Medical Research. Umesh Kumar. (Centre of Biomedical Research). 01/09/2015-30/08/2018. 10.150 €. Principal investigator. Conceptualization, study design, sample collection, sample preparation, Data recording and analysis, Manuscript preparation, project report
- 4 Project.** F.41-498/2012, Nematode constraint and their management in vegetable crops in western Uttar-Pradesh. University Grants Commission. Rajendra Singh. (Bareilly College Bareilly). 12/07/2012-11/07/2015. 10.902 €. Team member. Sample collection, Field work, Nematode extraction from infected plant root, culture, Isolation, Treatment, Manuscript writing, project report

C.5. Stays in public or private R&D centres

- 1** Leopold-Franzens Universität. Institute of Pharmacy. Austria. Innsbruck. 02/05/2023-30/06/2024. Post-doctoral.
- 2** National Bureau of Animal Genetic Resources. Department of Animal Biotechnology. India. Karnal. 24/12/2021-31/03/2023. Post-doctoral.